

Taylor Series for the F = ma Exam

A fluency-first toolkit: a handful of approximations you will use everywhere

This handout is not about *proving* anything. You do not need calculus, and you do not need to know what “convergence” means. The goal is fluency: by the end you should be able to look at $e^{0.03}$, $\sqrt{1.02}$, $\cos(0.1)$, or $1/\sqrt{1-v^2/c^2}$ and write down a good estimate in one line. These five facts and their uses appear over and over in introductory physics and on the F = ma exam. There is a total of **52** points. Problems marked **[A]** are a little harder.

1 The One Idea

Idea 1: Near a point, every curve looks simple

Zoom in on any smooth curve and it starts to look like a straight line; zoom out a little and it looks like a parabola. A **Taylor series** is just a recipe for writing a complicated function as

$$(\text{a constant}) + (\text{something}) \times x + (\text{something}) \times x^2 + \dots$$

where x is measured from the point you zoomed in on. When x is *small*, the later terms shrink fast and you can throw them away. That is the whole game.

Idea 2: “Small” means high powers are negligible

If $x = 0.1$, then $x^2 = 0.01$ and $x^3 = 0.001$. Each power is ten times smaller than the last. So if you only care about 1% accuracy, keeping the x term and dropping x^2 is completely safe. The smaller x is, the fewer terms you need. We write $\approx$ (“approximately equals”) to mean “equal once we drop the tiny leftover terms.”

- [3] Problem 1.** Let $x = 0.02$. Compute x , x^2 , and x^3 . If you want an answer good to about 0.1% (one part in a thousand), which of these terms can you safely ignore?

Solution. $x = 0.02$, $x^2 = 0.0004$, $x^3 = 0.000008$. One part in a thousand is 0.001. The $x^2 = 0.0004$ term is right at the edge and might matter; $x^3 = 8 \times 10^{-6}$ is far below 0.001 and is completely safe to drop. So you would keep through x^2 and ignore x^3 and beyond.

2 The Five You Must Memorize

Idea 3: The whole toolkit on one line each

For **small** x (and angles in **radians**):

$$(1+x)^n \approx 1+nx$$

$$e^x \approx 1+x$$

$$\ln(1+x) \approx x$$

$$\sin x \approx x$$

$$\cos x \approx 1 - \frac{1}{2}x^2$$

That is the entire handout in five boxes. Everything below is either a reason these are true, a graph of how good they are, or a place in physics where they save you. Notice sine keeps only *one* term (its next term is tiny, $-x^3/6$), while the others keep *two*.

Remark

Two cautions. (1) These hold only when x is small compared to 1. “ x small” for $\sin x \approx x$ literally means the angle in *radians* is small — $\sin(30^\circ)$ is *not* close to 30. (2) In $(1+x)^n$ the small quantity is x , not the whole base; you often have to massage a number into the form $1 + (\text{small})$ first.

Example 1: One-line estimates

$$e^{0.05} \approx 1 + 0.05 = 1.05 \quad (\text{true: } 1.0513), \quad \cos(0.1) \approx 1 - \frac{1}{2}(0.1)^2 = 0.995 \quad (\text{true: } 0.99500).$$

[5] **Problem 2.** Estimate each in one line using the boxed formulas, without a calculator:

- (a) $e^{0.01}$
- (b) $\sin(0.02)$
- (c) $\cos(0.2)$
- (d) $(1.03)^4$ (here $x = 0.03$, $n = 4$)
- (e) $\ln(1.04)$

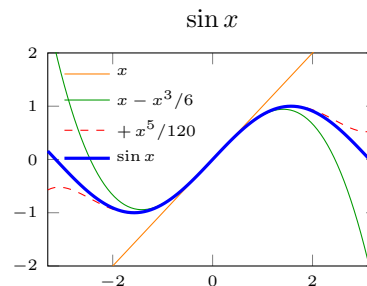
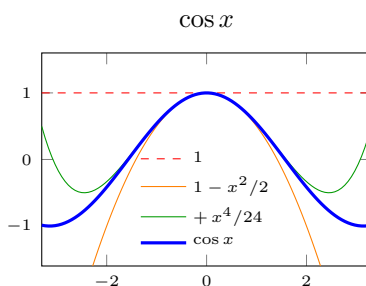
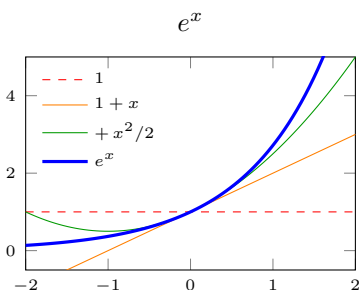
Solution. (a) $e^{0.01} \approx 1 + 0.01 = 1.01$. (b) $\sin(0.02) \approx 0.02$. (c) $\cos(0.2) \approx 1 - \frac{1}{2}(0.2)^2 = 1 - 0.02 = 0.98$. (d) $(1.03)^4 \approx 1 + 4(0.03) = 1.12$ (true 1.1255). (e) $\ln(1.04) \approx 0.04$.

3 More Terms, Better Fit**Idea 4: Each new term widens the range where the fit is good**

The full series for our functions are

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots, \quad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots, \quad \sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$$

You almost never need past the first two terms, because we only ever use these when x is small. But it is worth *seeing* what the extra terms buy you: one term matches the curve at a point, two terms match it over a small neighborhood, three terms over a wider one. The plots below show the true function (thick) against the 1-, 2-, and 3-term approximations.



Remark

Look at $\sin x \approx x$ near the middle of its plot: the straight line $y = x$ hugs the true sine curve all the way out to about $x = 0.5$ before peeling away. That is why the small-angle approximation is so good for pendulums — the swing angles are small.

- [4] **Problem 3.** Using the cosine series $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$, estimate $\cos(1)$ (that is one radian, about 57°) keeping (a) one term, (b) two terms, (c) three terms. The true value is 0.5403. Watch the estimate march toward it.

Solution. (a) 1. (b) $1 - \frac{1}{2} = 0.5$. (c) $1 - 0.5 + \frac{1}{24} = 0.5417$. Each term gets dramatically closer to 0.5403: the one-term answer is useless here because $x = 1$ is not small, but by three terms we are good to better than 0.3%.

4 Powers and Roots: $(1 + x)^n \approx 1 + nx$ for any n

Idea 5: n can be anything — negative, a fraction, whatever

The single formula $(1 + x)^n \approx 1 + nx$ does not care what n is. Three favorites:

$$\frac{1}{1+x} = (1+x)^{-1} \approx 1-x, \quad \sqrt{1+x} = (1+x)^{1/2} \approx 1+\frac{x}{2}, \quad \frac{1}{\sqrt{1+x}} = (1+x)^{-1/2} \approx 1-\frac{x}{2}.$$

A square root is nothing special: it is just the power $n = \frac{1}{2}$. Since $(1+x)^n \approx 1+nx$, we get $\sqrt{1+x} \approx 1 + \frac{1}{2}x$ for free.

Example 2: Estimating $\sqrt{65}$ by hand

$\sqrt{65}$ looks hard, but $65 = 64(1 + \frac{1}{64})$, and $\sqrt{64} = 8$ is easy. So

$$\sqrt{65} = 8\sqrt{1 + \frac{1}{64}} \approx 8\left(1 + \frac{1}{2} \cdot \frac{1}{64}\right) = 8\left(1 + \frac{1}{128}\right) = 8 + \frac{8}{128} = 8.0625.$$

The true value is 8.0623. We got four correct digits with mental arithmetic. The trick is always the same: **pull out the nearest easy value** so what is left is $1 + (\text{small})$.

Example 3: Why $\frac{1}{2}mv^2$ is the low-speed limit of relativity

You may have met Einstein's $E = mc^2$. The fuller statement of special relativity is that a mass m moving at speed v has total energy $E = \gamma mc^2$, where the **gamma factor** is

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = \left(1 - \frac{v^2}{c^2}\right)^{-1/2},$$

and c is the speed of light. We will just take this as a given fact. The number γ is always at least 1; at everyday speeds $v \ll c$ it is barely above 1, and it grows large only as v approaches c . The energy of motion (kinetic energy) is the part beyond the rest energy mc^2 , namely

$(\gamma - 1)mc^2$. Now Taylor-expand: with $x = -v^2/c^2$ small,

$$\gamma \approx 1 - \frac{1}{2} \left(-\frac{v^2}{c^2} \right) = 1 + \frac{1}{2} \frac{v^2}{c^2},$$

so the kinetic energy $(\gamma - 1)mc^2 \approx \frac{1}{2}mv^2$. The familiar $\frac{1}{2}mv^2$ you already know is just the first Taylor term of the exact relativistic energy.

[6] **Problem 4.** Estimate each by pulling out the nearest easy value, in one or two lines:

(a) $\sqrt{101}$ (use $100 \cdot (1 + \frac{1}{100})$)

(b) $\frac{1}{0.98}$

(c) $(1.01)^{10}$

(d) $\sqrt[3]{8.6}$ (use $8 \cdot (1 + \frac{0.6}{8})$, $n = \frac{1}{3}$)

(e) $\frac{1}{\sqrt{1.04}}$

(f) $\sqrt{0.96}$

Solution. (a) $\sqrt{101} = 10\sqrt{1+0.01} \approx 10(1+0.005) = 10.05$ (true 10.0499). (b) $\frac{1}{0.98} = (1 - 0.02)^{-1} \approx 1 + 0.02 = 1.02$ (true 1.0204). (c) $(1.01)^{10} \approx 1 + 10(0.01) = 1.10$ (true 1.1046). (d) $\sqrt[3]{8.6} = 2(1 + 0.075)^{1/3} \approx 2(1 + 0.025) = 2.05$ (true 2.0488). (e) $(1.04)^{-1/2} \approx 1 - 0.02 = 0.98$ (true 0.9806). (f) $\sqrt{0.96} = (1 - 0.04)^{1/2} \approx 1 - 0.02 = 0.98$ (true 0.9798).

[3] **Problem 5.** A pendulum's period is $T = 2\pi\sqrt{L/g}$. A clock's pendulum is lengthened by 0.5%. By what percentage does its period change? (Use $\sqrt{1+x} \approx 1 + \frac{x}{2}$.)

Solution. $T \propto \sqrt{L}$, so $T \rightarrow T\sqrt{1+0.005} \approx T(1+0.0025)$: the period grows by about 0.25%. (Half the fractional change in L , because of the square root.)

5 Logarithms: $\ln(1+x) \approx x$

Idea 6: Read it off from e^x

The logarithm undoes the exponential: $\ln(e^x) = x$. Now use $e^x \approx 1+x$. If we call $u = e^x - 1 \approx x$, then $1+u \approx e^x$, so

$$\ln(1+u) \approx \ln(e^x) = x \approx u.$$

In words: for small u , $\ln(1+u) \approx u$. The log of something just above 1 is about how far above 1 it is.

[4] **Problem 6.** Estimate, one line each:

(a) $\ln(1.1)$

- (b) $\ln(0.97)$ (write $0.97 = 1 + (-0.03)$)
- (c) $\ln(1.005)$
- (d) Sound level grows like $\ln$. If a sound's intensity rises by 2%, roughly how much does $\ln(\text{intensity})$ increase?

Solution. (a) $\ln(1.1) \approx 0.1$ (true 0.0953). (b) $\ln(0.97) \approx -0.03$ (true -0.0305). (c) $\ln(1.005) \approx 0.005$. (d) $\ln(1.02) \approx 0.02$, an increase of about 0.02.

6 Even and Odd Functions

Idea 7: Symmetry decides which powers can appear

A function is **even** if $f(-x) = f(x)$ — its graph is a mirror image across the vertical axis. It is **odd** if $f(-x) = -f(x)$ — its graph looks the same after a 180° rotation about the origin. Powers are the simplest examples:

$$x^0, x^2, x^4, \dots \text{ are even; } \quad x^1, x^3, x^5, \dots \text{ are odd.}$$

This forces the shape of a Taylor series:

- An **even** function's series uses *only even powers*. That is why $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$ has no x , no x^3 .
- An **odd** function's series uses *only odd powers*. That is why $\sin x = x - \frac{x^3}{6} + \dots$ has no constant, no x^2 .

Cosine is even, sine is odd. You can use this as a memory check: if you ever “derive” a $\cos x \approx 1 + ax$ with a stray linear term, you know it is wrong, because cosine is even.

- [4] **Problem 7.** Classify each as even, odd, or neither: (a) $x^2 + \cos x$, (b) $x \cos x$, (c) $\sin x + x^3$, (d) e^x .

Solution. (a) even (sum of two even things). (b) odd (odd $\times$ even = odd). (c) odd (sum of two odd things). (d) neither — its series $1 + x + \frac{x^2}{2} + \dots$ mixes even and odd powers.

- [3] **Problem 8.** A potential energy $U(x)$ near $x = 0$ is known to be an *even* function (the system looks the same if you flip $x \rightarrow -x$). Explain why its expansion must look like $U(x) \approx U_0 + \frac{1}{2}kx^2$ with no term linear in x — and why that means a mass placed there feels a spring-like (harmonic) force.

Solution. Even functions contain only even powers, so the x^1 term is forbidden; the first thing after the constant U_0 is an x^2 term, which we name $\frac{1}{2}kx^2$. A potential of the form $U_0 + \frac{1}{2}kx^2$ is exactly that of a spring, so the motion is harmonic. The symmetry alone guarantees no linear term, which is what kills any constant (non-restoring) force at the center.

7 Physics Applications

Idea 8: Small-angle pendulum

For a pendulum of length L swinging through angle θ , the restoring force along the arc is proportional to $\sin \theta$. Using $\sin \theta \approx \theta$,

$$\text{restoring force} \propto \sin \theta \approx \theta,$$

which is the defining feature of simple harmonic motion (force proportional to displacement). That single approximation is why a pendulum keeps near-constant time and why $T = 2\pi\sqrt{L/g}$ is independent of amplitude — *as long as the swing is small*. In energy language, the height gives $U = mgL(1 - \cos \theta) \approx mgL \cdot \frac{1}{2}\theta^2 = \frac{1}{2}(mgL)\theta^2$, a perfect parabola in θ : a spring with effective stiffness mgL .

Idea 9: Every potential is a spring near its minimum

This is the most important application on the whole list. Take *any* potential energy $U(x)$ with a minimum at $x = x_0$. Expand in the small displacement $\delta = x - x_0$:

$$U(x) \approx U(x_0) + (\text{slope at } x_0) \delta + \frac{1}{2}k \delta^2 + \dots$$

At a minimum the slope is zero, so the linear term *vanishes*. The first term that survives is the δ^2 term:

$$U(x) \approx U(x_0) + \frac{1}{2}k \delta^2$$

which is a spring. So a mass near *any* potential minimum oscillates harmonically with $\omega = \sqrt{k/m}$, where k is whatever number multiplies $\frac{1}{2}\delta^2$. Molecules, atoms in a crystal, a ball in a bowl, a pendulum — all of them are springs when you look closely enough at the bottom.

Example 4: Lennard–Jones: a molecule is a spring

Two atoms interact with the Lennard–Jones potential $U(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$, which has a minimum at some spacing r_0 . Near r_0 the linear term is absent (it is a minimum), so $U(r) \approx U(r_0) + \frac{1}{2}k(r - r_0)^2$. The molecule therefore vibrates like a tiny mass on a spring of stiffness k — exactly the model used for molecular vibration frequencies. The complicated r^{-12} and r^{-6} powers are irrelevant once you are close to the bottom; only the curvature k matters.

Idea 10: Gravity a little above the surface

Newton's law gives $g(h) = \frac{GM}{(R+h)^2}$ at height h above a planet of radius R . Factor out R :

$$g(h) = \frac{GM}{R^2} \frac{1}{(1+h/R)^2} = g_0 (1+h/R)^{-2} \approx g_0 \left(1 - \frac{2h}{R}\right).$$

So gravity weakens roughly *linearly* with height near the surface, losing a fraction $2h/R$ of

its strength. (The factor of 2 comes straight from $n = -2$.)

Idea 11: Radioactive decay over a short time

A sample decays as $N(t) = N_0 e^{-\lambda t}$. For times short compared with the lifetime ($\lambda t \ll 1$), use $e^{-\lambda t} \approx 1 - \lambda t$:

$$N(t) \approx N_0(1 - \lambda t),$$

so the fraction that has decayed is about $\lambda t = t/\tau$, growing linearly at first. This is why “activity” (decays per second) is nearly constant over short windows even though the true law is exponential.

- [3] **Problem 9.** At what height h (as a fraction of Earth’s radius $R \approx 6400$ km) does gravity drop by 1%? Use $g(h) \approx g_0(1 - 2h/R)$.

Solution. Set $2h/R = 0.01$, so $h/R = 0.005$ and $h \approx 0.005 \times 6400$ km = 32 km. A 1% drop in g takes only about 32 km of altitude.

- [3] **Problem 10.** A radioactive source has lifetime $\tau = 1/\lambda = 10$ hours. Using $e^{-\lambda t} \approx 1 - \lambda t$, estimate the fraction that has decayed after (a) 6 minutes and (b) 30 minutes. Why would this approximation be terrible for estimating the fraction left after 40 hours?

Solution. $\lambda = 1/(10$ h). (a) $t = 0.1$ h, fraction decayed $\approx \lambda t = 0.01$, i.e. 1%. (b) $t = 0.5$ h, fraction ≈ 0.05 , i.e. 5%. After 40 h, $\lambda t = 4$, which is not small at all — $1 - \lambda t = -3$ is nonsense (you cannot have a negative amount). The linear approximation only works while $\lambda t \ll 1$.

- [4] **Problem 11.** Show that the pendulum potential $U(\theta) = mgL(1 - \cos \theta)$ reduces to a spring for small swings, and identify the effective spring constant in the variable θ . Then argue in one sentence why the period does not depend on the amplitude of a small swing.

Solution. Using $\cos \theta \approx 1 - \frac{1}{2}\theta^2$, we get $U \approx mgL(1 - (1 - \frac{1}{2}\theta^2)) = \frac{1}{2}(mgL)\theta^2$, a spring with “stiffness” mgL in θ . Because the restoring effect is exactly proportional to the displacement θ (no higher powers survive), the oscillation frequency is set by the stiffness and inertia alone, not by how far you pull it back — so the period is amplitude-independent for small swings.

8 Propagation of Error

Idea 12: A power multiplies your percentage error by n

Suppose you measure a quantity with a small fractional (percentage) error ϵ , so your value is really $A(1 + \epsilon)$. If a result depends on it as a power, $Q = A^n$, then

$$Q = A^n(1 + \epsilon)^n \approx A^n(1 + n\epsilon).$$

The fractional error in Q is about $n\epsilon$: **powers scale percentage errors by the exponent**. Squaring doubles your error; a square root halves it; an inverse-square ($n = -2$) doubles it (the sign just says the error goes the other way).

Example 5: From a pendulum's period to g

You time a pendulum to get g from $T = 2\pi\sqrt{L/g}$, i.e. $g \propto T^{-2}$. Here $n = -2$, so a 1% error in your measured period T becomes a $|-2| \times 1\% = 2\%$ error in g . Timing matters twice as much as you might guess.

- [3] **Problem 12.** You measure the radius of a sphere to 1%. What is the resulting percentage error in (a) its surface area $\propto r^2$ and (b) its volume $\propto r^3$?

Solution. (a) $n = 2$: error $\approx 2 \times 1\% = 2\%$. (b) $n = 3$: error $\approx 3 \times 1\% = 3\%$. Higher powers amplify the measurement error in proportion to the exponent.

- [3] **Problem 13.** Kinetic energy is $K = \frac{1}{2}mv^2$. If a speed is known to 2%, to what percentage is the kinetic energy known (assume m is exact)? If instead you want K good to 1%, how precisely must you know v ?

Solution. $K \propto v^2$, so $n = 2$ and the error is $\approx 2 \times 2\% = 4\%$. To get K to 1% you need v to 0.5% (since $2 \times 0.5\% = 1\%$).

- [4] **Problem 14.** [A] A quantity is built as $Q = \frac{ab^2}{\sqrt{c}}$, where a, b, c are measured with small fractional errors $\epsilon_a, \epsilon_b, \epsilon_c$. Using $(1 + \epsilon)^n \approx 1 + n\epsilon$ on each factor, show that the fractional error in Q is approximately $\epsilon_a + 2\epsilon_b + \frac{1}{2}\epsilon_c$ in the worst case, and explain where each coefficient comes from.

Solution. Write each measured value as its true value times $(1 + \epsilon)$. Then

$$Q \propto a^1 b^2 c^{-1/2} \rightarrow (1 + \epsilon_a)^1 (1 + \epsilon_b)^2 (1 + \epsilon_c)^{-1/2} \approx (1 + \epsilon_a)(1 + 2\epsilon_b)(1 - \frac{1}{2}\epsilon_c).$$

Multiplying out and keeping only first-order terms (products of two small ϵ 's are negligible) gives $1 + \epsilon_a + 2\epsilon_b - \frac{1}{2}\epsilon_c$. So the fractional error coefficients are exactly the exponents: 1 for a , 2 for b (it is squared), and $\frac{1}{2}$ for c (it is under a square root). In the worst case the errors add in size, giving $\epsilon_a + 2\epsilon_b + \frac{1}{2}\epsilon_c$.

End of handout. If you remember only the five boxes in Section 2 and the trick of pulling out the nearest easy value, you already have what the F = ma exam asks of you.